

Volume with Whole Number Edges

Lesson 10-1

Name:

Date:

Class:

Key Vocabulary

Level 2 Standard

Picture first, then the word, then a plain-language meaning. Say each word out loud.



unit cubes

A box $3 \times 2 \times 4$ holds 24 unit cubes, so $V = 24$ cubic units

Volume

Write the definition:



length $\times$ width $\times$ height

A cereal box or shoe box — it has length, width, and height

Rectangular prism

Write the definition:

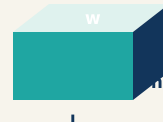


one cubic unit

A tiny cube that is $1 \text{ in} \times 1 \text{ in} \times 1 \text{ in} = 1 \text{ in}^3$ (one cubic inch)

Cubic units

Write the definition:



$V = l \times w \times h$; for a box $5 \times 3 \times 2$, volume = 30 cubic units

Length, width, height

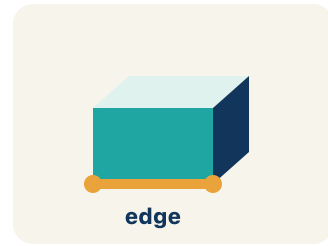
Write the definition:



A cross-shaped pattern of 6 rectangles folds into a rectangular box

Net

Write the definition:



A cube has 12 edges — 4 along the top, 4 along the bottom, 4 vertical

Edge

Write the definition:

Guided Notes

Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can find the volume of a rectangular prism with whole-number edges using $\text{length} \times \text{width} \times \text{height}$.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Volume

Rectangular prism

Cubic units

Length, width, height

Net

Edge

1 The amount of space inside a three-dimensional solid is its _____.

2 A solid with six rectangular faces, like a box, is a _____.

3 The unit used to measure volume, like cubic centimeters, is _____.

4 The three edge measurements of a rectangular prism are its _____.

5 A flat pattern that folds up into a solid is a _____.

6 A line segment where two faces of a solid meet is an _____.



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 I do — watch

Follow each step as your teacher solves it.

Problem: What is the volume of a rectangular prism with $l = 7$ in, $w = 3$ in, $h = 4$ in?

- A. 84 in^3
- B. 14 in^3
- C. 84 in^2
- D. 42 in^3

Step 1 $V = l \times w \times h = 7 \times 3 \times 4 = 84$ cubic inches.

 **Answer:** A. 84 in^3

 We do — together

Solve this one with your class using the same steps.

Problem: A time capsule box has a volume of 120 cm^3 . Its length is 10 cm and width is 4 cm. What is its height?

- A. 3 cm
- B. 6 cm
- C. 4 cm
- D. 12 cm

Step 1 _____

Step 2 _____

Answer: _____

 You do — your turn

Now try one on your own. Show every step.

Problem: Which unit is used for volume?

- A. Cubic inches (in^3)
- B. Square inches (in^2)
- C. Inches (in)
- D. Degrees ($^\circ$)

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. A time capsule box is 5 units long, 3 units wide, and 2 units tall. How many unit cubes fill it?

- A. 30 cubic units
- B. 10 cubic units
- C. 15 cubic units
- D. 25 cubic units

Show your work:

2. You stack unit cubes in a capsule: each layer has 6 cubes, and there are 4 layers. What is the volume?

- A. 24 cubic units
- B. 10 cubic units
- C. 12 cubic units
- D. 18 cubic units

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A box needs to hold exactly 60 cubic inches. Give three different sets of whole-number dimensions that work. Which set would make the box closest to a cube shape? Why might that matter?

Sentence starter: Option 1: $\underline{\hspace{1cm}} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}}$. Option 2: $\underline{\hspace{1cm}} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}}$. Option 3: $\underline{\hspace{1cm}} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}}$.

The $\underline{\hspace{1cm}}$ option is closest to a cube because $\underline{\hspace{1cm}}$. This matters because $\underline{\hspace{1cm}}$.

Show your work:

Reflect — Exit Ticket

A rectangular prism has $l = 11$ in, $w = 5$ in, $h = 4$ in. What is the volume?

- A. 220 in^3
- B. 55 in^3
- C. 220 in^2
- D. 200 in^3

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Volume
2. **Notes 2:** Rectangular prism
3. **Notes 3:** Cubic units
4. **Notes 4:** Length, width, height
5. **Notes 5:** Net
6. **Notes 6:** Edge
7. **We Try (notes):** A. 3 cm — $V = l \times w \times h \rightarrow 120 = 10 \times 4 \times h \rightarrow 120 = 40h \rightarrow h = 3 \text{ cm}$.
8. **You Try (notes):** A. Cubic inches (in^3) — *Volume measures 3D space, so it uses cubic units like in^3 . Square units (in^2) are for area, and inches are for length.*
9. **Try It 1:** A. 30 cubic units — $V = \text{length} \times \text{width} \times \text{height} = 5 \times 3 \times 2 = 30 \text{ cubic units}$.
10. **Try It 2:** A. 24 cubic units — *Each layer has 6 cubes and there are 4 layers, so $6 \times 4 = 24$ cubic units. This is the same as $\text{length} \times \text{width} \times \text{height}$ with one factor being the number of layers.*
11. **Exit Ticket:** A. 220 in^3 — $V = 11 \times 5 \times 4 = 220 \text{ cubic inches}$. Remember: volume uses cubic units (in^3).